Big Data Analytics: Mini Project 2
Project Title: Spark Case Study

Assignment Weight	Team Size	Due Date
8 Points	2 Students	25th April 2026

1. Project Objective

The goal of this project is to **analyze large datasets using Spark SQL, DataFrames, and RDDs**, and to compare their performance and execution plans. Students will learn how Spark's **Catalyst Optimizer** and **Tungsten Engine** improve query execution, and why Structured APIs are preferred over RDDs.

2. Problem Definition

Each team must select a dataset from open sources (Kaggle, UCI ML Repository, government portals, or other reliable sources).

- The dataset must support a **unique analytical problem** (*not just descriptive statistics*).
- Teams must define a **research-style question** (e.g., prediction, anomaly detection, trend analysis).
    - Example:
        - *Healthcare Data:* Study patient readmission patterns and resource allocation.
        - *Online Retail*: Study customer segmentation and sales trends.
        - *Web Logs*: Investigate user activity and bandwidth usage.

3. Dataset Requirements

- The dataset must be **large enough** to demonstrate Spark's distributed processing.
- **Size:** $\geq$ 1 million records.
- **Attributes:** $\geq$ 8 columns, mixed data types (numeric, categorical, string, timestamp).
- **Format:** CSV, JSON, Parquet, or semi-structured logs.
- **Justification:** Teams must explain why the dataset is suitable for distributed processing and what insights it enables.

4. Query Requirements

Each team must design and execute **at least 10 queries**, including:

1. Filtering with complex conditions.
2. Aggregations (SUM, AVG, COUNT, MAX/MIN).
3. Grouping by multiple attributes.
4. Sorting and ranking results.
5. Window functions (moving averages, cumulative sums, ranking).
6. Nested queries or subqueries.
7. **Advanced optimization cases:** broadcast join vs. sort-merge join, partition pruning, caching.

5. Implementation Requirements

- Each query must be implemented in **three ways**:
  - **RDD API** .
  - **DataFrame API** .
  - **Spark SQL**.
- Teams must compare:
  - **Execution plans** (.explain(True) for DataFrames/Spark SQL).
  - **Performance metrics** (execution time, shuffle size, memory usage).
  - **Scalability tests:** run queries with different cluster sizes or partitioning strategies.

6. Optimization & Analysis

- For each query, teams must:
  - Show the **initial logical plan**.
  - Show the **optimized logical plan** (Catalyst transformations).
  - Show the **physical plan** (execution strategy).
- At least two queries must demonstrate **join optimization** (broadcast join vs. sort-merge join).
- Compare **RDD vs. DataFrame vs. SQL performance**.
- Evaluate the impact of **caching, partitioning, and file formats (CSV vs. Parquet)**

7. Deliverables

Each team must submit the following:

1- a comprehensive project report including:

- **Problem Statement**
  - Dataset chosen and unique analytical problem defined.
- **Dataset Description**
  - Source, size, schema, and justification for choice.
- **Queries Implemented**
  - SQL code, DataFrame code, and RDD code for each query.
  - Screenshots of .explain() outputs for DataFrame/SQL queries.
- **Execution Plan Comparison**
  - Logical plan, optimized plan, and physical plan for each query.
  - Highlight differences between RDD, DataFrame, and SQL execution.
- **Performance Comparison Table**
  - Execution time, shuffle size, memory usage for each query.
  - Identify which API performed best and why.
- **Final Insights**
  - Business or analytical conclusions drawn from the dataset.
  - Lessons learned about Spark optimization.

2- Code Submission

  - All source code files organized by API type.
  - Clear folder structure with instructions to run.
  - Include scripts for cluster configuration and scalability tests.

8. Evaluation Criteria

- **Dataset Choice (15%)** – Large, diverse, and justified.
- **Query Complexity (25%)** – Meets minimum requirements and demonstrates advanced features.
- **Optimization Analysis (25%)** – Clear comparison of plans and performance.
- **Documentation Quality (20%)** – Well-structured report with screenshots and tables.
- **Insights & Conclusions (15%)** – Meaningful analysis of results.

Due Date:
Saturday, 25th April 2026 – 11:59 PM
Important: Late submissions will not be accepted and will receive zero marks.